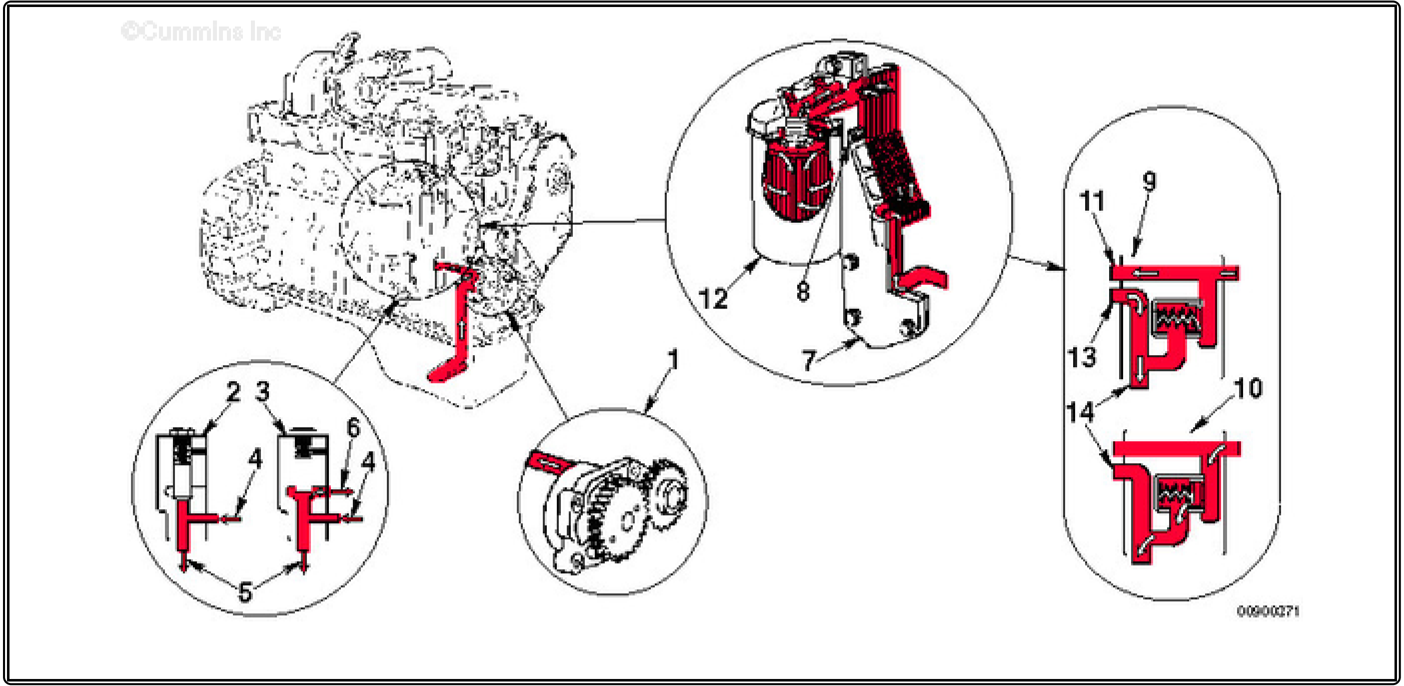
Trainings

Flow Diagram



1. Gerotor lubricating oil pump
2. Pressure regulating valve closed
3. Pressure regulating valve open
4. From lubricating oil pump
5. To lubricating oil cooler
6. To lubricating oil pan
7. Lubricating oil cooler
8. Filter bypass valve
9. Filter bypass valve closed
10. Filter bypass valve open
11. To lubricating oil filter
12. Full-flow lubricating oil filter
13. From lubricating oil filter
14. Main lubricating oil rifle.

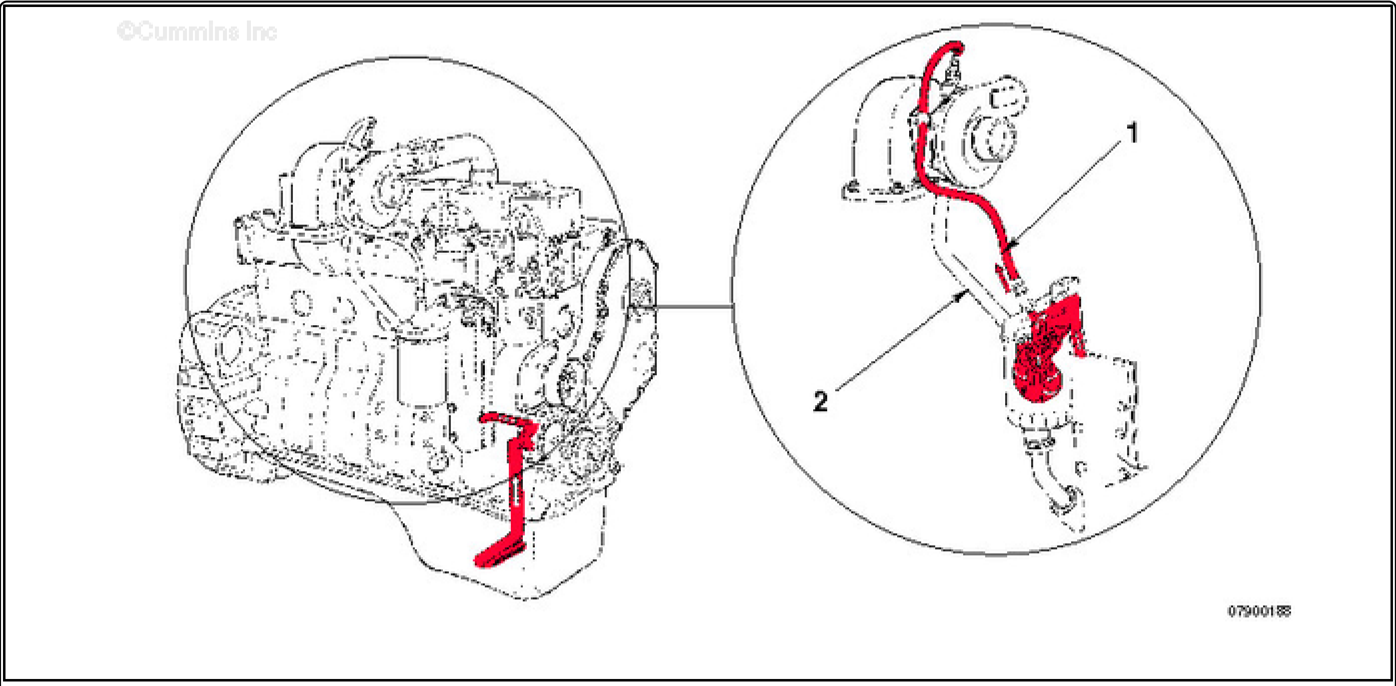
Lubricating Oil Pump

The engine uses a gerotor-type lubricating oil pump (1). The machined cavity in the block is the same for all engines. A wider gerotor is used in the six-cylinder engine to increase the lubricating pump capacity. Consequently, the four-cylinder and six-cylinder lubricating pumps are **not** interchangeable.

Pressure-Regulating Valve

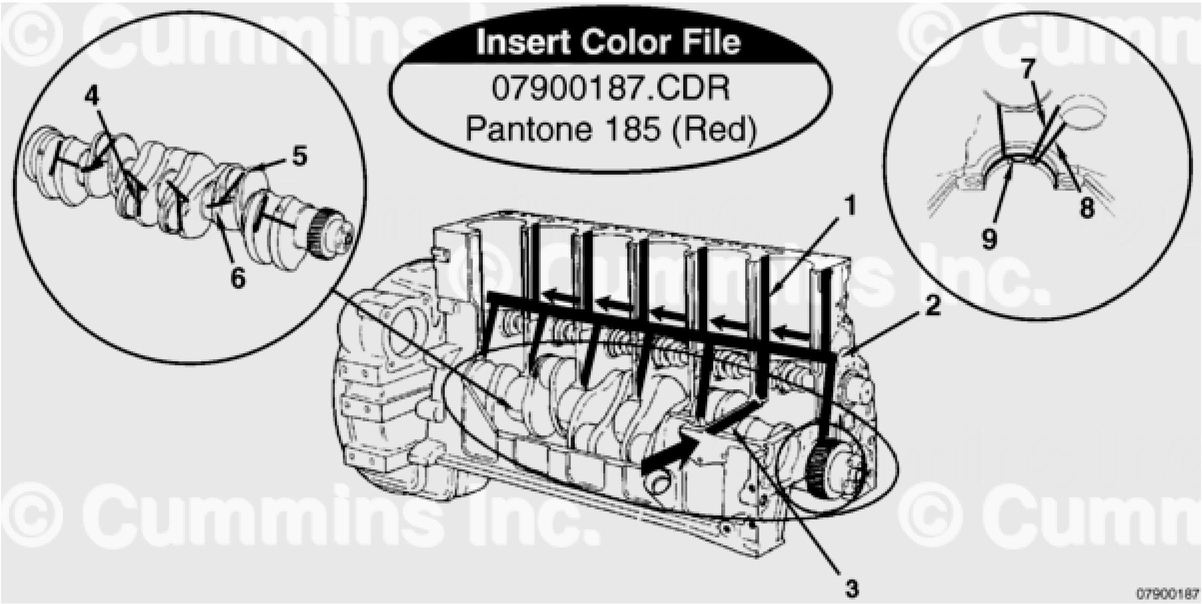
The pressure-regulating valve (2) is designed to keep the lubricating oil pressure from exceeding 449 kPa [65 psi]. When the lubricating oil pressure from the pump is greater than 449 kPa [65 psi], the valve opens, uncovering the dump port, so part of the lubricating oil is routed to the oil pan. The minimum lubricating oil pressure limit is the same for the four-cylinder and the six-cylinder engine. Because of manufacturing tolerances of the components and the oil passages, the lubricating oil pressure can differ as much as 69 kPa [10 psi] between engines.

Lubrication for the Turbocharger



1. Oil supply
2. Oil drain.

The turbocharger receives filtered, cooled, and pressurized lubricating oil through a supply line from the filter head. A drain line connected to the bottom of the turbocharger housing returns the lubricating oil to the lubricating oil pan through a fitting in the cylinder block.



Lubrication for the Power Components

1. To valve train
2. Main oil rifle
3. From oil cooler
4. Connecting rod journal
5. To connecting rod bearing
6. Crankshaft, main journal
7. From main oil rifle
8. To camshaft
9. To piston cooling nozzle.

The main bearings and the valve train are lubricated by pressurized oil directly from the main oil rifle. The other power components, connecting rods, pistons, and camshaft receive pressurized oil indirectly from the main oil rifle.

The drillings in the crankshaft supply oil to the connecting rod bearings. The oil is supplied to the camshaft journals through drillings in the main bearing saddle. Smaller drillings in the main bearing saddle supply oil to the piston cooling nozzles. The spray from the nozzles also provides lubrication for the piston pins.

The Number 1 main bearing saddle does **not** contain a piston cooling nozzle. Cylinder Number 1 receives the lubricating and cooling spray from the nozzle located in the Number 2 bearing saddle. Cylinder Number 2 receives the spray from the Number 3 bearing saddle, etc.

Lubrication for the valve train is supplied through separate drillings in the cylinder block. The oil flows through the drillings and across the oil transfer slot in the cylinder head gasket. From the transfer slot, the oil flows around the outside diameter at the cylinder head capscrew, across a slot in the bottom of the rocker lever support, and up a vertical drilling in the support. From these drillings, oil flows through drillings in the rocker lever shaft to lubricate the rocker levers. Oil flows through a drilling in the rocker levers to fill a channel cast into the top of the levers. The oil from the channel lubricates the valve stems, push tubes, and tappets.

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